NEF ABUNDANCE: SURFACE BEHAVIOR

This note studies the smooth surface case of the nef-abundance proof in *Numerical Semiampleness of Nef Adjoint Divisors*.

In this study note, we work over $\mathbb{C}$.

Let X be a smooth projective surface with $K_X \sim 0$, and let D be a nef Cartier divisor with $D^2 = 0$. In Section 2, we additionally assume $\mathrm{n}(X, D) = 2$.

For a nef divisor D, its nef reduction is an almost holomorphic rational map $f : X \dashrightarrow Y$ with connected fibres such that D is numerically trivial on the general compact fibre and $D \cdot C > 0$ for every curve through a very general point not contracted by f. Write $\mathrm{n}(X, D) = \dim Y$. A very general point lies outside a countable union of proper closed subsets. For a Cartier divisor H, write $E \in |H|$ if E is an effective Cartier divisor linearly equivalent to H. If g is a local equation of E at x, define

$$\mathrm{mult}_x(E) = \max\{a \geq 0 : g \in \mathfrak{m}_x^a\},$$

where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$. This is independent of multiplication of g by a unit. A nonzero section $s \in H^0(X, \mathcal{O}_X(H))$ gives its zero divisor $E = (s)_0 \in |H|$, and

$$\mathrm{mult}_x(E) = \mathrm{ord}_x(s).$$

Conversely every $E \in |H|$ arises this way, with the section determined up to a nonzero scalar.

1. Preliminary lemmas

Lemma 1.1. *Let X be a smooth projective surface, let D be a nef Cartier divisor with $D^2 = 0$, and let A be an ample Cartier divisor. For every $\epsilon > 0$, there are positive integers $r_n \to \infty$ such that*

$$P_n := r_n(D + A/n), \qquad P_n^2 \to \epsilon^2.$$

Proof. Put

$$u_n = \sqrt{(D + A/n)^2} = \sqrt{2D \cdot A/n + A^2/n^2} > 0.$$

Then $u_n \to 0$. For sufficiently large n, choose r_n to be the nearest integer to ϵ/u_n. This integer is positive and

$$|r_n u_n - \epsilon| \leq u_n/2 \to 0, \qquad r_n \to \infty.$$

Set $r_n = 1$ for the remaining finitely many indices. Hence $P_n^2 = (r_n u_n)^2 \to \epsilon^2$. □

Lemma 1.2. *Let X be a smooth projective surface and let H be a nef Cartier divisor. Let $r \in \mathbb{R}_{\geq 0}$ and let $\Lambda \subseteq |H|$ be a nonempty linear subsystem. Suppose y is an isolated point of its base locus and $\mathrm{mult}_y(E) \geq r$ for every $E \in \Lambda$. Then*

$$r^2 \leq H^2.$$

Proof. Let F be the fixed divisorial part of Λ. Choose a general $E_1 \in \Lambda$ and write $E_1 = E_1' + F$. Each coefficient along F has its generic minimal value, so E_1' is effective and contains no component of F. Since y is an isolated base point, $y \notin \mathrm{Supp}(F)$, and hence

$$\mathrm{mult}_y(E_1') = \mathrm{mult}_y(E_1) \geq r.$$

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The residual system $\Lambda - F$ has no fixed curves. Thus a general $E_2 \in \Lambda$ has no component in common with E_1'. Consequently E_1', E_2 meet properly and $\mathrm{mult}_y(E_2) \geq r$. The local intersection inequality and nefness of H give

$$\begin{aligned} r^2 &\leq \mathrm{mult}_y(E_1')\,\mathrm{mult}_y(E_2) \leq i_y(E_1', E_2) \leq E_1' \cdot E_2 \\ &= H \cdot E_1' \leq H \cdot E_1 = H^2. \end{aligned}$$

□

In particular, if every member has multiplicity at least $r > \sqrt{H^2}$ at y, the nonempty system has a base curve through y.

Lemma 1.3. *Let X be a smooth projective surface, H a Cartier divisor, $\Lambda \subseteq |H|$ a fixed nonempty linear subsystem, and $r \in \mathbb{R}$. Suppose that there exists a Zariski dense subset $S \subseteq X$ such that for every $x \in S$ there is $E_x \in \Lambda$ with $\mathrm{mult}_x(E_x) > r$. Then for every $x \in X$ there exists a divisor $E_x \in \Lambda$ with $\mathrm{mult}_x(E_x) > r$.*

Proof. We may assume $r \geq 0$. Put

$$Z = \{(x, E) \in X \times \Lambda : \mathrm{mult}_x(E) > r\}.$$

Put $m = \dim \Lambda$. In local coordinates u, v on X and homogeneous parameter coordinates $[t_0 : \cdots : t_m]$ on Λ, the universal effective Cartier divisor $\mathcal{H}$ is given by

$$f(u, v, t) = \sum_{i=0}^{m} t_i f_i(u, v) = 0,$$

where the f_i represent a basis of the defining section space. Let $c_{i,ab}(x)$ be the regular coefficient functions of $u^a v^b$ in the local expansion of f_i in parameters centered at x. Then Z is defined locally on X by the homogeneous equations

$$\sum_{i=0}^{m} t_i c_{i,ab}(x) = 0 \qquad (a, b \geq 0,\ a + b \leq \lfloor r \rfloor),$$

since $\mathrm{mult}_x(E_t) > r$ exactly when all these low-degree coefficients vanish. Hence Z is closed in $X \times \Lambda$.

Since Λ is projective, the projection $p : X \times \Lambda \to X$ is proper, so $p(Z)$ is closed. The hypothesis says that $p(Z)$ contains S, and hence $p(Z) = X$. □

Lemma 1.4. *Let X be a smooth projective surface and A an ample Cartier divisor. Suppose that for every $y \in X$ there is $E \in |A|$ with $\mathrm{mult}_y(E) > 4\sqrt{A^2}$. Then, through a general point of X, there is an integral curve C such that*

$$0 < A \cdot C \leq 2\sqrt{A^2}.$$

Proof. We next use Lemma 1.2 to force a base curve through the marked point. Choose a sufficiently large positive integer k with $k\sqrt{A^2} \geq 4$. Multiplying the divisors in the hypothesis by k gives members of $|kA|$ of multiplicity strictly greater than $4k\sqrt{A^2}$. For $y \in X$, put

$$\Lambda_i(y) = \{E \in |kA| : \mathrm{mult}_y(E) \geq \lceil (2i+3)k\sqrt{A^2}/2 \rceil\} \quad (i = 0, 1, 2).$$

All three systems are nonempty for very general y, by the hypothesis, and their base loci are proper. If y were an isolated point of $\mathrm{Bs}\,\Lambda_0(y)$, Lemma 1.2 with $r = h_0 = \lceil 3k\sqrt{A^2}/2 \rceil$ would give

$$1 \leq (k/h_0)^2 A^2 \leq \tfrac{4}{9} < 1,$$

a contradiction. Thus $\mathrm{Bs}\,\Lambda_0(y)$ contains a curve through y.

We construct a family of base curves with a dominant marked-point map. On a dense marked-point open, jet evaluation has constant rank. The systems $\Lambda_i(y)$ are projectivizations of the corresponding section-kernel vector bundles V_i. We have

$$\Lambda_2(y) \subseteq \Lambda_1(y) \subseteq \Lambda_0(y), \qquad \operatorname{Bs}\Lambda_0(y) \subseteq \operatorname{Bs}\Lambda_1(y) \subseteq \operatorname{Bs}\Lambda_2(y).$$

A curve component of $\operatorname{Bs}\Lambda_0(y)$ remains a component of $\operatorname{Bs}\Lambda_1(y)$, since a proper closed subset of a surface has dimension at most one. Track such a component over the generic marked point. Its finitely many possible components are defined after a finite parameter-field extension. Spreading out and shrinking gives a smooth integral parameter surface T, a dominant generically finite marked-point map $y : T \to X$, and a family of integral curves C_t common to $\operatorname{Bs}\Lambda_0(y(t))$ and $\operatorname{Bs}\Lambda_1(y(t))$. Its incidence

$$\Gamma = \{(t, z) : z \in C_t\} \subset T \times X$$

dominates X, because $y(t) \in C_t$.

We compare parameter derivatives with the lower jet kernel. Set $h = \lceil k\sqrt{A^2}/2 \rceil$. Choose local defining-section representatives from a frame of V_1, the section bundle underlying Λ_1. Each has the form

$$s(t, z) = \sum_\alpha c_\alpha(t) s_\alpha(z),$$

with s_α fixed global sections of kA. Parameter derivatives with z fixed are again global sections at each fixed parameter. In coordinates centered at the moving mark $y(t)$, s starts in order $\lceil 5k\sqrt{A^2}/2 \rceil$. Each parameter derivative lowers this order by at most one. Thus every derivative of order $a < h$ belongs to the section space $V_0(t)$ underlying $\Lambda_0(y(t))$, since

$$\lceil 5k\sqrt{A^2}/2 \rceil - \lceil 3k\sqrt{A^2}/2 \rceil \geq k\sqrt{A^2} - 1 \geq h - 1 \geq a.$$

Every such derivative vanishes along Γ.

We deduce multiplicity at least h along the incidence. At a general smooth point of Γ, the dominant map $\Gamma \to X$ is submersive in characteristic zero. Parameter directions therefore span its normal space in $T \times X$. Since Γ has codimension one, choose a parameter coordinate t_1 transverse to it. In completed local coordinates the formal implicit-function theorem writes

$$\Gamma : \ t_1 = F(t_2, z), \qquad u = t_1 - F(t_2, z).$$

Holding t_2, z fixed gives $\partial/\partial t_1 = \partial/\partial u$ to every order. Use a frame of kA depending only on z. The vanishing of all parameter derivatives of order less than h along Γ forces all coefficients of u^a with $a < h$ to vanish. Hence $s \in \mathcal{I}_\Gamma^h$ in the completed local ring, and faithful flatness of completion gives the same containment in the algebraic local ring. Apply this to a finite frame and shrink to a dense algebraic open of Γ.

We restrict to the fibres and intersect with A to obtain the degree bound. For a general t, the fibre of the incidence is reduced and agrees locally with C_t. Restricting the containment shows that every defining section of $\Lambda_1(y(t))$ lies in $\mathcal{I}_{C_t}^h$ at the generic point of C_t. Thus every $E \in \Lambda_1(y(t))$ satisfies $E \geq hC_t$. Since $E \sim kA$ and A is ample,

$$0 < A \cdot C_t \leq (k/h)A^2 \leq 2\sqrt{A^2}.$$

We obtain the assertion on a nonempty open subset of X. Shrink T to a nonempty open T° where the fibres are integral and the bound holds. By Chevalley's theorem, the image of the dominant map $y : T^\circ \to X$ contains a nonempty open subset $U \subset X$. Every $x \in U$ lies on such a curve C_t, since $y(t) \in C_t$. This proves the assertion for general points.

□

Higher-dimensional form of the same argument. The following statements retain the normalization, full jet kernels, parameter differentiation, and intersection argument of the preceding lemmas. The dimension is now n; the surface statements are their specializations. In higher dimension the first output is a moving subvariety, whose dimension need not be one.

Lemma 1.5. *Let X be a smooth projective variety of dimension n, let D be nef with $D^n = 0$, and let A be ample. For every $\epsilon > 0$ and every sequence of positive rational numbers $t \to 0$, there are positive integers $r_t \to \infty$ such that*

$$P_t = r_t(D + tA), \qquad P_t^n \longrightarrow \epsilon^n.$$

If D, A are Cartier, P_t is an ample $\mathbb{Q}$-Cartier divisor; its Cartier multiples are chosen after this normalization.

Proof. Put $a_t = ((D + tA)^n)^{1/n}$. Then $a_t > 0$ and $a_t \to 0$. Choose r_t to be the nearest positive integer to ϵ/a_t. For sufficiently small t, $|r_t a_t - \epsilon| \le a_t/2$, which proves the assertions. □

Lemma 1.6. *Let X be a smooth projective n-fold, let H be a nef Cartier divisor, and let $\Lambda \subseteq |H|$ be a nonempty linear system with base ideal $\mathfrak{b}$. Let V be an integral component of its base locus, of codimension $c \ge 1$. If*

$$\mathfrak{b}\mathcal{O}_{X,\eta_V} \subseteq \mathfrak{m}_{\eta_V}^h \qquad (h \in \mathbb{Z}_{>0}),$$

then

$$h^c\, H^{n-c} \cdot V \le H^n.$$

In particular, if x is an isolated base point and every member has multiplicity at least a real number $r \ge 0$ at x, then $r^n \le H^n$.

Proof. Intersect successively with general members of Λ. Before each next cut, remove the components already contained in the base locus. Each remaining component is cut properly, and the intersection cycle is effective. Its H-degree equals the H-degree of the preceding cycle; discarding effective components can only decrease this degree, since H is nef. Starting with $[X]$, the H-degree of the codimension-c cycle is therefore at most H^n.

At η_V, the localized base ideal is primary for the maximal ideal. No discarded component of smaller codimension contains η_V, because V is an irreducible component of the base locus. The first c general equations $s_1, \dots, s_c$ form a parameter ideal I in the regular local ring $R = \mathcal{O}_{X,\eta_V}$. The coefficient of V in the intersection cycle is $\operatorname{length}(R/I)$. Since $I \subseteq \mathfrak{m}^h$, monotonicity of Hilbert–Samuel multiplicity gives

$$\operatorname{length}(R/I) = e(I, R) \ge e(\mathfrak{m}^h, R) = h^c.$$

Here the first equality uses that a parameter sequence in R is regular, and the last equality uses regularity of R. This proves the inequality. For an isolated point take $c = n$ and $h = \max\{1, \lceil r \rceil\}$. □

Lemma 1.7. *Lemma 1.3 holds for a smooth projective variety of any dimension.*

Proof. Use local parameters $u_1, \dots, u_n$ and homogeneous parameters $[t_0 : \cdots : t_m]$ on the fixed linear system. Write the universal equation as $f = \sum_i t_i f_i$, and denote its coefficient functions by $c_{i,\alpha}(x)$, where $\alpha \in \mathbb{Z}^n_{\ge 0}$. For $r \ge 0$, the incidence Z is defined locally by

$$\sum_i t_i c_{i,\alpha}(x) = 0 \qquad (|\alpha| \le \lfloor r \rfloor).$$

These coefficients are the finite local jet coefficients. Thus Z is closed, and its proper projection has closed image. An image containing a Zariski dense subset equals X. For $r < 0$ any member of the nonempty system suffices. □

Proposition 1.8. *Let X be a smooth projective n-fold over $\mathbb{C}$, with $n \geq 2$, and let A be an ample $\mathbb{Q}$-Cartier divisor. Put $\rho = (A^n)^{1/n}$. Suppose that, for some positive integer k_* with k_*A Cartier, a Zariski dense set of points x admits $E_x \in |k_*A|$ with $\operatorname{mult}_x(E_x) > 4k_*\rho$. Then there is a covering family of integral subvarieties V of one fixed dimension d, with $1 \leq d < n$, such that*

$$0 < A^d \cdot V \leq n^{n-d}(A^n)^{d/n}.$$

In particular, through every point of a nonempty open subset there is such a V. For $n = 2$ this is Lemma 1.4.

Proof. Lemma 1.7 makes the multiplicity condition valid at every point. Taking powers of the defining sections allows k to be arbitrarily large and divisible by k_*. Put

$$\delta = \frac{2\rho}{n}, \qquad \tau_i = \frac{3}{2}\rho + i\delta \quad (0 \leq i \leq n), \qquad q_i = \lceil k\tau_i \rceil.$$

Choose k so that $k\delta \geq 4$ and $k\rho/2 > 1$. All full kernel spaces

$$V_i(x) = H^0(X, \mathcal{O}_X(kA) \otimes \mathfrak{m}_x^{q_i})$$

are nonzero, because $\tau_n = 7\rho/2 < 4\rho$. Write $B_i(x)$ for their base loci. They are proper and nested. The point x cannot be isolated in $B_0(x)$: otherwise Lemma 1.6 would give $q_0^n \leq k^n A^n$, whereas $q_0 \geq 3k\rho/2$.

Begin with a positive-dimensional component through x in $B_0(x)$, and at each next level choose a component containing it. Strict containment increases dimension. Among the n adjacent pairs, some pair therefore has a common component, since all dimensions lie between 1 and $n-1$. On a dense marked-point open all jet ranks are constant. After a finite extension of the generic parameter field, spread out the chosen components and shrink. This gives a smooth integral parameter variety T, a dominant generically finite map $x : T \to X$, a fixed index i, and integral fibres V_t common to $B_i(x(t))$ and $B_{i+1}(x(t))$. The incidence $\Gamma \subset T \times X$ dominates X, since $x(t) \in V_t$.

Set $h = \lceil k\delta/2 \rceil$. A local frame of the higher kernel has defining sections $s(t,z) = \sum_\alpha a_\alpha(t)s_\alpha(z)$, where the s_α are fixed global sections of kA. Every parameter derivative of total order $a < h$, with z fixed, is still a global section and has multiplicity at least $q_{i+1} - a$ at $x(t)$. Moreover

$$q_{i+1} - q_i \geq k\delta - 1 \geq h - 1.$$

It therefore belongs to the full lower kernel and vanishes along Γ. This is the parameter-differentiation mechanism of Ein–Küchle–Lazarsfeld, Proposition 2.3; see the original statement.

Put $c = n - d$. At a general smooth point of Γ, its dominant projection to X is smooth, so parameter directions span its normal space. Choose c transverse parameter coordinates. In completed local coordinates the incidence has equations $u_j = t_j - F_j(t_{c+1}, \ldots, t_n, z)$, $1 \leq j \leq c$. Holding the other parameters and z fixed identifies $\partial/\partial t_j$ with $\partial/\partial u_j$. Vanishing of all derivatives of total order less than h implies $s \in \mathcal{I}_\Gamma^h$; here characteristic zero is used. Faithful flatness of completion gives the same algebraic containment. Apply it to the finite frame. On a dense incidence open, restriction to a general fibre gives ordinary-power containment in $\mathcal{I}_{V_t}^h$ at the generic point of V_t.

The higher base ideal is primary there, since V_t is its component. Lemma 1.6, applied to $H = kA$, gives

$$h^{n-d}k^d A^d \cdot V_t \leq k^n A^n, \qquad A^d \cdot V_t \leq (k/h)^{n-d}A^n \leq (n/\rho)^{n-d}A^n.$$

Ampleness gives positivity. The marked-point image of a suitable open in T contains a nonempty open subset of X, as required. □

In dimension three the alternatives are a curve with $A \cdot C \leq 9(A^3)^{1/3}$, or a surface with $A^2 \cdot V \leq 3(A^3)^{2/3}$. The surface alternative must be treated before asserting a covering family of curves. These statements are proof drafts for checking; the source's later subadjunction and bounded-curve arguments require their own hypothesis audit.

2. Jet estimates

Lemma 2.1. *Let X be a smooth projective K3 surface, and let D be a nef Cartier divisor with $\mathrm{n}(X, D) = 2$. Fix an ample Cartier divisor A and $\epsilon \in \mathbb{Q}_{>0}$ with*

$$(14\epsilon)^2 < \tfrac{1}{4}, \qquad 80\epsilon < \tfrac{3}{4}.$$

Let the positive integer n tend to infinity, and choose positive integers r_n such that

$$P_n = r_n(D + A/n), \qquad r_n \to \infty, \qquad P_n^2 \to \epsilon^2.$$

There exist a positive integer N and proper closed subsets $Z_i \subsetneq X$ $(i \geq 1)$ such that every point

$$x \in X \setminus \bigcup_{i \geq 1} Z_i$$

satisfies, simultaneously for every $n \geq N$, every positive integer k with kP_n Cartier, and every effective divisor $E \in |kP_n|$,

$$\mathrm{mult}_x(E) \leq 4k\sqrt{P_n^2}.$$

Proof. We use Lemma 1.4.

Suppose that the asserted bound fails generically for arbitrarily large indices n. Fix such an index and write

$$P = P_n, \qquad r = r_n, \qquad \rho = \sqrt{P^2}.$$

The ranks of jet evaluation maps are algebraic rank conditions. Outside a countable union of proper closed subsets, a violation for some k therefore gives a nonzero complete jet kernel at the general point. Put $H = kP$. Lemma 1.3 with $\Lambda = |H|$ and $r = 4\sqrt{H^2}$ gives a divisor of multiplicity greater than $4\sqrt{H^2}$ at every point. Apply Lemma 1.4 to the ample Cartier divisor H. Its nonempty open locus meets the very-general nef-reduction locus. Choose an integral curve C through a point in this intersection with

$$0 < H \cdot C \leq 2\sqrt{H^2} = 2k\rho, \qquad 0 < P \cdot C \leq 2\rho.$$

Choose this point sufficiently very general for the nef reduction of D. Since $\mathrm{n}(X, D) = 2$ and D is Cartier, $D \cdot C > 0$ implies $D \cdot C \geq 1$. Moreover $P - rD = (r/n)A$ is nef. Hence

$$r \leq rD \cdot C \leq P \cdot C \leq 2\rho.$$

This is impossible for arbitrarily large n, because $r_n \to \infty$ and $\rho_n = \sqrt{P_n^2} \to \epsilon$. No higher-dimensional bounded-curve argument is needed: the proper positive-dimensional base component on a surface is already a curve.

Consequently some N, depending only on the fixed data and the chosen sequence, works at the general point for every $n \geq N$. Take the union of the exceptional rank loci for the countably many indices, Cartier multiples and integral jet orders, together with the exceptional loci used for the nef reduction. This gives the single countable union in the statement. □

Here N depends on the fixed data X, D, A, ϵ and the chosen sequence (r_n), independently of x. A single very general point works for every $n \geq N$: take the union of the exceptional countable unions for the countably many indices n.

Proposition 2.2. *Let X, D satisfy the common hypotheses above. Fix an ample Cartier divisor A and $\epsilon \in \mathbb{Q}_{>0}$ with $(14\epsilon)^2 < 1/4$ and $80\epsilon < 3/4$. Let $n \to \infty$ through positive integers and choose positive integers r_n such that $P_n = r_n(D + A/n)$, $r_n \to \infty$, and $P_n^2 \to \epsilon^2$. Let $p_i : X \times X \to X$ be the projections, and put*

$$Z = \mathbb{P}(p_1^*\mathcal{O}_X(D) \oplus p_2^*\mathcal{O}_X(D)) \xrightarrow{\pi} X \times X, \qquad \xi = c_1(\mathcal{O}_Z(1)),$$

using the convention $\pi_*\mathcal{O}_Z(a) = \mathrm{Sym}^a(p_1^*\mathcal{O}_X(D) \oplus p_2^*\mathcal{O}_X(D))$ for $a \geq 0$. Write $D_i = \pi^* p_i^* D$ and $P_{n,i} = \pi^* p_i^* P_n$. The two natural sections of π are the axes, of classes $\xi - D_1$ and $\xi - D_2$. For a suitable fixed positive integer q and all sufficiently large n, the ample rational divisor

$$R_n = P_{n,1} + P_{n,2} + q\xi$$

has a smooth point $z \in Z$ outside both axes such that

$$\varepsilon(R_n; z) := \inf_{C \ni z} \frac{R_n \cdot C}{\mathrm{mult}_z C} > 6,$$

where C ranges over integral curves through z. For any prescribed dense open subsets $U_1, U_2 \subseteq X$, the point may be chosen with $\pi(z) \in U_1 \times U_2$.

Proposition 2.2 is recorded from OpenAI, *Numerical Semiampleness of Nef Adjoint Divisors* (October 3, 2026), Sections 3–6; its proof is not supplied or audited here. Here $\dim Z = 5$. The order of choices is ϵ, then a fixed q, then sufficiently large n.

Corollary 2.3. *Let X, D satisfy the common hypotheses above. Fix an ample Cartier divisor A and $\epsilon \in \mathbb{Q}_{>0}$ with $(14\epsilon)^2 < 1/4$ and $80\epsilon < 3/4$. Let $n \to \infty$ through positive integers and choose positive integers r_n such that $P_n = r_n(D + A/n)$, $r_n \to \infty$, and $P_n^2 \to \epsilon^2$. With $p_i : X \times X \to X$ the projections, set*

$$Z = \mathbb{P}(p_1^*\mathcal{O}_X(D) \oplus p_2^*\mathcal{O}_X(D)) \xrightarrow{\pi} X \times X, \qquad \xi = c_1(\mathcal{O}_Z(1)),$$

with the symmetric-power convention for $\mathcal{O}_Z(1)$. Fix a suitable positive integer q and a sufficiently large index n for which

$$R_n = \pi^*(p_1^* P_n + p_2^* P_n) + q\xi \quad \text{and} \quad \varepsilon(R_n; z) > 6$$

at a smooth point $z \in Z$ outside the two natural sections of π. Here $\varepsilon(R_n; z) = \inf_{C \ni z}(R_n \cdot C)/\mathrm{mult}_z C$. Then, for sufficiently large divisible k_0 with $k_0 R_n$ and $k_0 P_n$ Cartier, the evaluation map

$$H^0(Z, \mathcal{O}_Z(k_0 R_n)) \longrightarrow \mathcal{O}_Z(k_0 R_n)_z \otimes_{\mathcal{O}_{Z,z}} \mathcal{O}_{Z,z}/\mathfrak{m}_z^{6k_0+1}$$

is surjective, where $\mathfrak{m}_z$ is the maximal ideal of $\mathcal{O}_{Z,z}$. Thus the sections generate jets through order $6k_0$.

Proof. Choose $a \in \mathbb{Q}$ with $6 < a < \varepsilon(R_n; z)$. On the blowup $b : \widetilde{Z} \to Z$ at z, with exceptional divisor G, the divisor $b^* R_n - aG$ is ample. For sufficiently large divisible k_0, Serre vanishing gives

$$H^1(\widetilde{Z}, \mathcal{O}_{\widetilde{Z}}(k_0 b^* R_n - k_0 a G)) = 0.$$

The restriction sequence gives surjectivity onto the local quotient modulo $\mathfrak{m}_z^{k_0 a}$. Taking k_0 also so that $k_0(a - 6) \geq 1$ yields $k_0 a \geq 6k_0 + 1$, and hence the stated map. □

After a local trivialization and smooth coordinates $u_1, \ldots, u_5$ at z, every prescribed Taylor polynomial spanned by $u_1^{a_1} \cdots u_5^{a_5}$ with $\sum a_i \leq 6k_0$ is realized. This is surjectivity for all prescribed Taylor data, rather than only existence of one section with large vanishing order. The surface arguments above supply the two auxiliary inputs for Lemma 2.1. The full higher-dimensional theorem and the remaining Frobenius comparison are outside the verification in this note.

Source. OpenAI, *Numerical Semiampleness of Nef Adjoint Divisors*, Sections 3–6, especially the two auxiliary estimates in Section 4 and the paragraph "The finite jet system" in Section 6. The local manuscript sources were checked directly, including the companion *Log abundance in characteristic zero*, Section 7, for the auxiliary arguments.

3. The finite jet system and its first Frobenius consequence

Fix q, n, z as in Corollary 2.3, and put $P = P_n$ and $R = P_1 + P_2 + q\xi$. Choose k_0 once and for all as in that corollary. Since $\dim Z = 5$, a local frame at z identifies its jet target with

$$J_{6k_0}(z) = \mathbb{C}[u_1, \dots, u_5]/(u_1, \dots, u_5)^{6k_0+1}, \qquad \dim_{\mathbb{C}} J_{6k_0}(z) = \binom{6k_0+5}{5}.$$

Thus a finite collection of sections suffices: choose preimages of a basis of this quotient. All these sections, q, n, z and k_0 are fixed over $\mathbb{C}$ before any reduction to positive characteristic.

The projective-bundle convention and the Künneth formula give

$$H^0(Z, \mathcal{O}_Z(k_0R)) \simeq \bigoplus_{j=0}^{qk_0} H^0(X, \mathcal{O}_X(k_0P + jD)) \otimes H^0(X, \mathcal{O}_X(k_0P + (qk_0 - j)D)).$$

Let $\pi(z) = (x_1, x_2)$. Trivialize the two summands near this pair and choose the fibre coordinate t so that the weight-j monomial is t^j. Since z lies off both axes, $t(z) = c \neq 0$. For a tensor-product basis, a local section is $f_{j,a}(u_1, u_2)g_{j,b}(u_3, u_4)t^j$. The column of its jet matrix consists of the coefficients

$$[u_1^{\alpha_1} \cdots u_4^{\alpha_4} v^{\alpha_5}]\big(f_{j,a}g_{j,b}(c+v)^j\big), \qquad |\alpha| \leq 6k_0,$$

where $v = t - c$. Corollary 2.3 says precisely that this finite matrix has row rank $\binom{6k_0+5}{5}$. This is a simultaneous prescription of all Taylor coefficients, not merely a section with one high vanishing order.

Once a model carrying the chosen sections and jet frames has been spread over an integral finitely generated $\mathbb{Z}$-algebra, the full-rank condition is preserved by inverting one nonzero maximal minor. This elementary rank statement only concerns the fixed finite system. It does not by itself supply the other geometric or uniform estimates in the Frobenius argument.

Lemma 3.1. *Let Y be a smooth variety over an algebraically closed field, let $y \in Y$, and let L be a line bundle. Suppose a finite space of global sections generates jets through order m at y. For every positive integer N, the products of N of these sections generate jets through order Nm for $L^{\otimes N}$.*

Proof. Choose regular parameters $u_1, \dots, u_d$ and a local frame for L. For each $\alpha \in \mathbb{Z}^d_{\geq 0}$ with $|\alpha| \leq m$, choose a section whose expansion is u^α modulo $\mathfrak{m}_y^{m+1}$. For any β with $|\beta| \leq Nm$, distribute its $|\beta|$ individual variable factors into N groups of size at most m. This writes $\beta = \alpha^{(1)} + \cdots + \alpha^{(N)}$ with $|\alpha^{(i)}| \leq m$; empty groups are allowed. The product of the corresponding sections has expansion

$$u^\beta + \text{terms of total degree greater than } |\beta|.$$

Order the monomial basis of the quotient by increasing total degree. The selected products have a triangular coefficient matrix with diagonal entries one. They therefore span the full quotient modulo $\mathfrak{m}_y^{Nm+1}$. This proof uses Taylor coefficients and does not divide by factorials; it works in every characteristic. □

Lemma 3.2. *Let Y be a smooth projective surface over an algebraically closed field of characteristic p, let D be Cartier, and let P be $\mathbb{Q}$-Cartier. Let $q, k_0 > 0$ be integers with k_0P Cartier. On*

$$Z_Y = \mathbb{P}(\mathcal{O}_Y(D)_1 \oplus \mathcal{O}_Y(D)_2), \qquad R = P_1 + P_2 + q\xi,$$

suppose a fixed finite system of sections of k_0R generates jets through order $6k_0$ at a point z off both axes. Choose a Cartier divisor T so that every $T + aD$, for $0 \le a \le (k_0 - 1)q$, has sections nonzero at both base coordinates of z. Put

$$N_p = \lfloor p/k_0 \rfloor, \qquad B_p = k_0N_pP + T.$$

If $p \ge 6k_0$, sections of $pq\xi + B_{p,1} + B_{p,2}$ surject onto the local quotient by the p-th powers of five regular parameters at z.

Proof. Lemma 3.1 gives jets through order $6k_0N_p$ for N_pk_0R. Set $d_p = (p - k_0N_p)q$, so $0 \le d_p \le (k_0 - 1)q$. Choose sections of T and $T + d_pD$ nonzero at the first and second base coordinates, respectively. Their tensor product, in the weight-zero summand of the bundle formula, gives a section of $d_p\xi + T_1 + T_2$ nonzero at z: its fibre monomial is the d_p-th power of the second coordinate. Multiplication by this section is invertible on every local jet quotient and changes N_pk_0R into $pq\xi + B_{p,1} + B_{p,2}$.

Moreover,

$$6k_0N_p \ge 6(p - k_0 + 1) \ge 5(p - 1) \qquad (p \ge 6k_0).$$

Every monomial not in $(u_1^p, \dots, u_5^p)$ has total degree at most $5(p - 1)$. Hence

$$\mathfrak{m}_z^{6k_0N_p+1} \subseteq (u_1^p, \dots, u_5^p).$$

Ordinary jet surjectivity therefore implies the asserted surjectivity onto the Frobenius quotient. □

Corollary 3.3. *Under Lemma 3.2, let $F : Y \to Y'$ be relative Frobenius and let D' be the twist of D. Put $\mathcal{E} = F_*\mathcal{O}_Y(B_p)$. Then $\operatorname{rk}\mathcal{E} = p^2$, and the natural evaluation map on $Y' \times Y'$,*

$$\bigoplus_{a+b=q} H^0(Y, \mathcal{O}_Y(B_p + paD)) \otimes H^0(Y, \mathcal{O}_Y(B_p + pbD)) \otimes \mathcal{O}(-aD'_1 - bD'_2) \longrightarrow \mathcal{E} \boxtimes \mathcal{E},$$

has generic rank p^4. Here a, b are nonnegative integers.

Proof. Smoothness makes F finite flat of degree p^2. Let A_* be the tensor product of the two local Frobenius-fibre algebras at the base coordinates of z. Each has dimension p^2, so $\dim A_* = p^4$. With the above fibre coordinate and $c = t(z) \ne 0$, the five-parameter Frobenius quotient is

$$A_*[t]/(t^p - c^p).$$

It is free over the possibly nonreduced algebra A_* with basis $1, t, \dots, t^{p-1}$. Project to the coefficient of 1. For $j = pa + b$ with $0 \le b < p$, reduction of t^j gives $c^{pa}t^b$. Thus precisely the weights $j = pa$, $0 \le a \le q$, survive. Their coefficients belong to

$$H^0(Y, B_p + paD) \otimes H^0(Y, B_p + p(q - a)D).$$

The projection formula identifies these with the values of the displayed evaluation map in local frames; each multiplier c^{pa} is nonzero. Lemma 3.2 says the whole quotient is spanned, so its projection spans A_*. The evaluation has rank p^4 at this base pair, hence has generic rank p^4. No reducedness of A_* is used. □

The preceding arguments explain the first rank calculation in Section 6 of the source paper. They are conditional on the fixed jet system, whose construction here still uses the unaudited Proposition 2.2. The remaining comparison needs a separately fixed cotangent-slope flag, a movable-curve witness on a point blowup, and estimates uniform in p for diagonal ranks and determinant vanishing. These inputs are not proved in this note. In particular, no proof that D has an effective numerical representative, or that the branch $D^2 = 0$, $\mathrm{n}(X, D) = 2$ is impossible, is asserted here.

Disclosure. This AI-assisted note is intended for joint learning, discussion, and checking. Its correctness has not been independently verified. No claim of novelty, priority, or authorship credit is made; unproved inputs are identified above.